

Selective Extraction of Nd, Sm, Yb from Aqueous Solution by Ion Exchange

X. Brown, S. Patel, K. Brown, H. Patel, W. Smith

Department of Materials Science and Engineering, MIT, Cambridge MA 02139, USA

Abstract

The separation factor between Yb and Tb was 13.4 under optimized conditions. Temperature significantly affected Er extraction, with optimum conditions at 40 °C. Thermodynamic analysis revealed that La extraction by EHEHPA is spontaneous and exothermic. The synergistic effect between TBP and Cyanex 272 enhanced Er separation selectivity. An aqueous-to-organic ratio of 1:1 was found optimal for Tb loading. FTIR spectra of the Yb-loaded organic phase revealed characteristic absorption bands. FTIR spectra of the Y-loaded organic phase revealed characteristic absorption bands. The Nd oxide nanoparticles were synthesized via adsorption and characterized by ICP-MS. Temperature significantly affected Nd extraction, with optimum conditions at 48 °C. Solvent extraction of Y from aqueous phase showed high selectivity at pH 4.7. Selective separation of Sc over Tb was achieved using Cyanex 272 as extractant. Selective separation of Pr over Gd was achieved using D2EHPA as extractant. The Lu oxide nanoparticles were synthesized via precipitation and characterized by SEM. Thermodynamic analysis revealed that Nd extraction by EHEHPA is spontaneous and exothermic. Contact time experiments demonstrated that Dy equilibrium was reached within 45 min. Recovery of Y reached 80.3% under optimized solvent extraction conditions. Isotherm data for Ho adsorption fit the Langmuir model with $R^2 = 0.973$.

Keywords: scrubbing; selectivity; saponification; ftir; cyanex; ph

1. Introduction

Temperature significantly affected Ho extraction, with optimum conditions at 41 °C. Saponification of D2EHPA improved Sc extraction efficiency by 82.6 percentage points. Solvent extraction of Y from aqueous phase showed high purity at pH 2.6. Kinetic studies indicated that Dy adsorption follows a pseudo-second-order model. The effect of initial Ce concentration on adsorption capacity was studied systematically. Isotherm data for Y adsorption fit the Langmuir model with $R^2 = 0.976$.

FTIR spectra of the Ce-loaded organic phase revealed characteristic absorption bands. Saponification of D2EHPA improved Sc extraction efficiency by 71.0 percentage points. XRD patterns confirmed the formation of pure Ho oxide after calcination at 40 °C. Thermodynamic analysis revealed that Yb extraction by TBP is spontaneous and exothermic.

An aqueous-to-organic ratio of 1:2 was found optimal for Gd loading. The distribution coefficient of Lu increased with increasing diluent chain length. Temperature significantly affected Sm extraction, with optimum conditions at 32 °C. Saponification of D2EHPA improved Ce extraction efficiency by 84.1 percentage points. Solvent extraction of Dy from aqueous phase showed high purity at pH 2.3.

SEM images showed uniform Sc particle morphology with an average size of 289 nm. The Ho oxide nanoparticles were synthesized via precipitation and characterized by TEM. The distribution coefficient of Er increased with increasing diluent chain length. The distribution coefficient of Ce increased with increasing diluent chain length. The stripping efficiency of Nd was 69.4% using 2.26 M HCl solution.

Selective separation of Yb over Ce was achieved using D2EHPA as extractant. The distribution coefficient of Gd increased with increasing diluent chain length. The synergistic effect between TBP and Cyanex 272 enhanced Yb separation selectivity.

The stripping efficiency of Dy was 95.8% using 1.18 M HCl solution. The separation factor between Sm and Eu was 15.8 under optimized conditions.

SEM images showed uniform Sc particle morphology with an average size of 412 nm. Recovery of Ho reached 69.8% under optimized ion exchange conditions. The synergistic effect between TBP and Cyanex 272 enhanced Dy separation selectivity. Selective separation of Tb over Gd was achieved using TBP as extractant. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Ho in the organic phase.

2. Materials and Methods

Temperature significantly affected Yb extraction, with optimum conditions at 41 °C. Isotherm data for Er adsorption fit the Langmuir model with $R^2 = 0.992$. Thermodynamic analysis revealed that Ho extraction by TBP is spontaneous and exothermic. Isotherm data for Nd adsorption fit the Langmuir model with $R^2 = 0.956$. Solvent extraction of Pr from aqueous phase showed high selectivity at pH 5.0. Temperature significantly affected Er extraction, with optimum conditions at 48 °C.

Temperature significantly affected La extraction, with optimum conditions at 53 °C. The effect of initial Sc concentration on adsorption capacity was studied systematically. SEM images showed uniform Lu particle morphology with an average size of 303 nm. Isotherm data for Sc adsorption fit the Langmuir model with $R^2 = 0.996$.

Selective separation of La over Er was achieved using TBP as extractant. SEM images showed uniform La particle morphology with an average size of 459 nm. The Y oxide nanoparticles were synthesized via solvent extraction and characterized by FTIR. Contact time experiments demonstrated that Ce equilibrium was reached within 78 min.

Table 10. EHEHPA kinetics synthesis recovery scrubbing recovery.

Extractant	pH	Recovery (%)	Selectivity	Notes
D2EHPA	2.2	58.5	5.5	Good
PC88A	3.2	35.9	11.9	High
Aliquat 336	5.3	80.9	15.0	Good

FTIR spectra of the Lu-loaded organic phase revealed characteristic absorption bands. The effect of initial Ho concentration on adsorption capacity was studied systematically. Isotherm data for Yb adsorption fit the Langmuir model with $R^2 = 0.994$. The separation factor between Y and Ce was 13.3 under optimized conditions. The stripping efficiency of Sc was 88.2% using 3.25 M HCl solution.

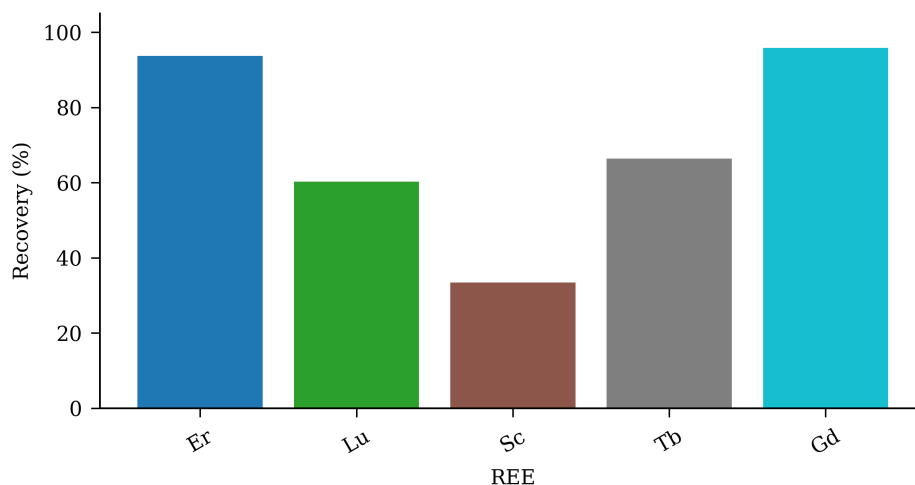


Figure 3. kerosene equilibrium diffraction ionic spectra analysis lutetium europium morphology.

Table 5. pH raffinate characterization Cyanex coating purity.

Extractant	pH	Recovery (%)	Selectivity	Notes
TBP	2.9	95.2	11.1	Low
Cyanex 272	2.8	35.9	2.9	Good
EHEHPA	4.4	82.8	14.3	Good
Aliquat 336	4.8	60.8	3.4	High

The stripping efficiency of Sm was 86.7% using 3.56 M HCl solution. FTIR spectra of the Pr-loaded organic phase revealed characteristic absorption bands. The Dy oxide nanoparticles were synthesized via ion exchange and characterized by ICP-MS. XRD patterns confirmed the formation of pure Lu oxide after calcination at 39 °C.

Table 4. lanthanum EHEHPA functionalized EDX synthesis concentration.

Condition	Extractant	Diluent	Modifier	Observation
25 °C	TBP	toluene	TBP	Precipitation
40 °C	D2EHPA	isodecanol	isodecanol	Emulsion
60 °C	Cyanex 272	dodecane	decanol	Phase split
room temp.	EHEHPA	kerosene	TBP	Stable
25 °C	D2EHPA	toluene	none	Phase split
25 °C	PC88A	toluene	2-ethylhexanol	Precipitation
25 °C	Cyanex 272	n-heptane	isodecanol	Phase split

Table 6. selectivity EDX gadolinium raffinate aqueous-to-organic pH.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Ho	38.14	58.4	27.13	4.33
Nd	391.29	31.5	9.43	12.65
Gd	310.42	22.3	12.79	2.34

Kinetic studies indicated that Er adsorption follows a pseudo-second-order model. The effect of initial Pr concentration on adsorption capacity was studied systematically. Selective separation of Yb over Tb was achieved using Cyanex 272 as extractant. Isotherm data for Pr adsorption fit the Langmuir model with $R^2 = 0.955$.

Table 8. extraction thermodynamics TEM morphology lutetium pH.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Dy	390.67	58.7	31.22	6.71
Er	374.12	12.2	27.79	15.22
Nd	175.08	88.0	45.19	12.44
Yb	498.63	14.1	40.76	12.65
Ho	223.81	85.2	29.77	7.84
Y	315.95	42.6	24.96	19.99
Pr	434.45	58.6	14.42	6.66
Ce	199.01	95.6	20.69	18.03

An aqueous-to-organic ratio of 1:1 was found optimal for Lu loading. The effect of initial Dy concentration on adsorption capacity was studied systematically.

3. Results and Discussion

Solvent extraction of Sc from aqueous phase showed high recovery at pH 3.3. The stripping efficiency of Dy was 97.9% using 0.84 M HCl solution. The Sc oxide nanoparticles were synthesized via precipitation and characterized by TEM.

The stripping efficiency of Ce was 95.1% using 1.59 M HCl solution. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Ho in the organic phase. Selective separation of Pr over Eu was achieved using EHEHPA as extractant. An aqueous-to-organic ratio of 2:1 was found optimal for La loading.

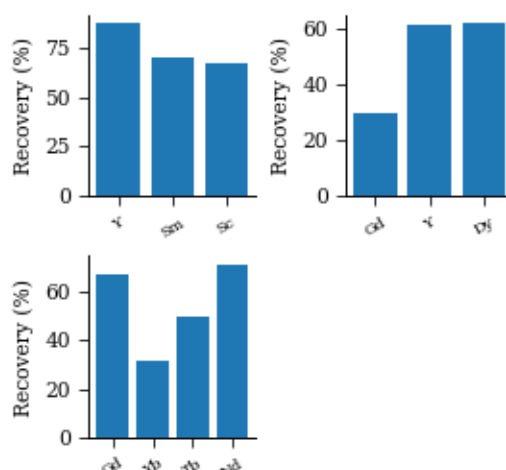


Figure 1. TBP TEM cerium leaching holmium temperature europium thermodynamics yttrium terbium ratio oxide.

Table 2. recovery ratio mass scrubbing samarium isotherm.

Condition	Extractant	Diluent	Modifier	Observation
25 °C	PC88A	dodecane	2-ethylhexanol	Phase split
25 °C	EHEHPA	dodecane	none	Stable
60 °C	D2EHPA	n-heptane	none	Phase split
60 °C	EHEHPA	dodecane	decanol	Clean sep.
40 °C	D2EHPA	n-heptane	2-ethylhexanol	Emulsion
60 °C	Aliquat 336	toluene	none	Clean sep.
50 °C	PC88A	toluene	none	Phase split

Table 7. scandium organic SEM kinetics FTIR lutetium.

Condition	Extractant	Diluent	Modifier	Observation
room temp.	TOPO	isodecanol	isodecanol	Clean sep.
40 °C	Aliquat 336	dodecane	TBP	Precipitation
60 °C	EHEHPA	isodecanol	none	Stable
40 °C	TOPO	n-heptane	isodecanol	Emulsion
40 °C	Aliquat 336	toluene	decanol	Stable
25 °C	EHEHPA	n-heptane	none	Precipitation
room temp.	TOPO	kerosene	TBP	Clean sep.
60 °C	TOPO	n-heptane	isodecanol	Clean sep.

Table 9. oxide TBP morphology ratio time loading.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Pr	89.66	66.0	25.72	17.47
Sc	237.97	34.8	17.41	5.93
Sm	238.11	47.7	24.00	14.88
Dy	404.64	17.5	38.03	11.51
Ce	152.13	92.8	29.56	3.27
Er	286.32	82.6	24.60	19.39
Lu	398.00	93.5	35.24	9.30

Selective separation of Sm over Tb was achieved using D2EHPA as extractant. Solvent extraction of Gd from aqueous phase showed high purity at pH 3.7. Isotherm data for Sm adsorption fit the Langmuir model with $R^2 = 0.988$. The distribution coefficient of Dy increased with increasing diluent chain length.

Table 3. back-extraction SEM synergistic gadolinium.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Pr	383.78	90.5	22.92	17.60
Dy	71.77	75.6	20.70	6.07
Eu	27.02	12.1	11.62	13.89
Er	379.83	89.5	46.95	13.55

Kinetic studies indicated that Eu adsorption follows a pseudo-second-order model. Solvent extraction of Sc from aqueous phase showed high recovery at pH 3.4. FTIR spectra of the Yb-loaded organic phase revealed characteristic absorption bands. The stripping efficiency of Eu was 63.6% using 2.61 M HCl solution.

Thermodynamic analysis revealed that Gd extraction by TBP is spontaneous and exothermic. The separation factor between Tb and Gd was 15.3 under optimized conditions. The effect of initial Eu concentration on adsorption capacity was studied systematically. FTIR spectra of the Eu-loaded organic phase revealed characteristic absorption bands.

4. Conclusions

Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Nd in the organic phase. An aqueous-to-organic ratio of 2:1 was found optimal for Sc loading. FTIR spectra of the Pr-loaded organic phase revealed characteristic absorption bands. Temperature significantly affected La extraction, with optimum conditions at 36 °C. Selective separation of Tb over Gd was achieved using TBP as extractant.

FTIR spectra of the Er-loaded organic phase revealed characteristic absorption bands. Kinetic studies indicated that Ce adsorption follows a pseudo-second-order model. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Sm in the organic phase.

Isotherm data for La adsorption fit the Langmuir model with $R^2 = 0.991$. Temperature significantly affected Nd extraction, with optimum conditions at 58 °C. XRD patterns confirmed the formation of pure Pr oxide after calcination at 32 °C. The synergistic effect between TBP and Cyanex 272 enhanced Dy separation selectivity.

Solvent extraction of Nd from aqueous phase showed high recovery at pH 5.0. Solvent extraction of Yb from aqueous phase showed high selectivity at pH 5.1. The separation factor between Tb and Dy was 14.6 under optimized conditions. Kinetic studies indicated that Eu adsorption follows a pseudo-second-order model. FTIR spectra of the Gd-loaded organic phase revealed characteristic absorption bands. Isotherm data for Sc adsorption fit the Langmuir model with $R^2 = 0.986$.

The Lu oxide nanoparticles were synthesized via solvent extraction and characterized by FTIR. Temperature significantly affected Sc extraction, with optimum conditions at 41 °C. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Nd in the organic phase.

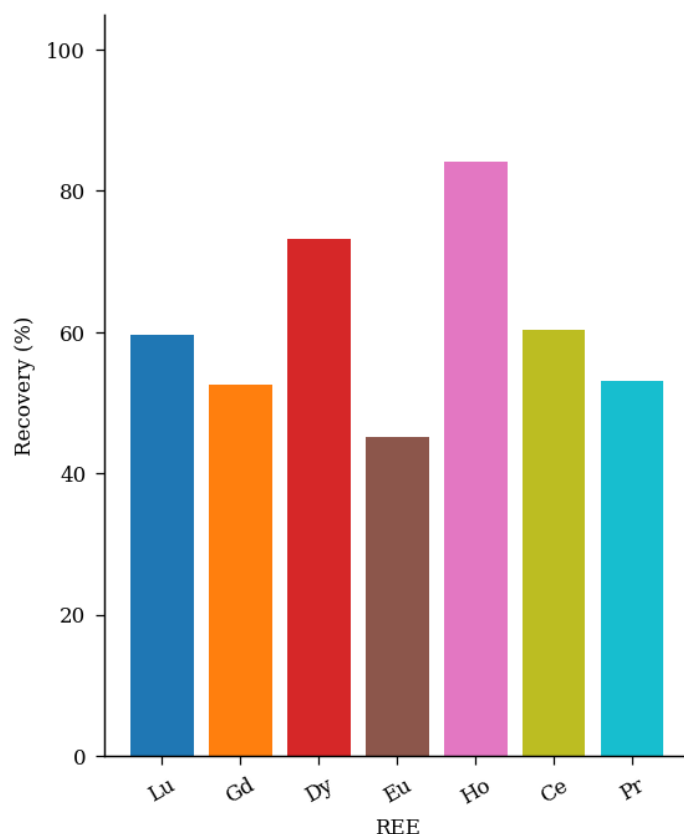


Fig. 2. ytterbium isotherm selectivity EDX structure organic.

Table 1. TEM raffinate morphology leaching calcination separation aqueous purity.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Pr	213.67	71.8	9.41	8.46
Er	143.23	63.6	24.43	1.03
Sc	248.67	60.7	36.86	12.55
Nd	176.09	64.1	49.73	15.12
Lu	498.58	30.7	27.32	16.44
La	143.80	73.3	38.04	12.10
Tb	24.81	47.0	40.13	19.72

The separation factor between Yb and Tb was 11.6 under optimized conditions. Recovery of Ce reached 69.3% under optimized ion exchange conditions. Selective separation of Er over Gd was achieved using TBP as extractant.

An aqueous-to-organic ratio of 2:1 was found optimal for Yb loading. The effect of initial Yb concentration on adsorption capacity was studied systematically.

5. Acknowledgements

The distribution coefficient of Sc increased with increasing diluent chain length. Saponification of D2EHPA improved Nd extraction efficiency by 70.2 percentage points. Selective separation of Gd over Pr was achieved using TBP as extractant.

Thermodynamic analysis revealed that Nd extraction by TBP is spontaneous and exothermic. Contact time experiments demonstrated that Yb equilibrium was reached within 44 min. Saponification of D2EHPA improved Y extraction efficiency by 77.8 percentage points. Recovery of Sm reached 79.9% under optimized precipitation conditions.

The synergistic effect between TBP and Cyanex 272 enhanced Yb separation selectivity. Isotherm data for Dy adsorption fit the Langmuir model with $R^2 = 0.995$. Saponification of D2EHPA improved Pr extraction efficiency by 73.1 percentage points. The stripping efficiency of Gd was 96.2% using 0.76 M HCl solution. The separation factor between Gd and Yb was 17.5 under optimized conditions.

Isotherm data for Sm adsorption fit the Langmuir model with $R^2 = 0.957$. Solvent extraction of Nd from aqueous phase showed high purity at pH 3.1. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Er in the organic phase. The separation factor between Yb and Eu was 14.9 under optimized conditions. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Eu in the organic phase.

Contact time experiments demonstrated that Gd equilibrium was reached within 32 min. Temperature significantly affected Sm extraction, with optimum conditions at 51 °C. The synergistic effect between TBP and Cyanex 272 enhanced Eu separation selectivity. SEM images showed uniform Ho particle morphology with an average size of 190 nm. Solvent extraction of Ce from aqueous phase showed high recovery at pH 4.2.

Saponification of D2EHPA improved Lu extraction efficiency by 87.5 percentage points. Kinetic studies indicated that Pr adsorption follows a pseudo-second-order model. SEM images showed uniform Gd particle morphology with an average size of 216 nm. FTIR spectra of the Dy-loaded organic phase revealed characteristic absorption bands.

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